

NLP for Phishing Detection resources

NLP for Phishing Detection resources shows how to build a simple phishing email detector using Supervised Machine Learning + NLP:

- Input: labeled emails (legit vs phishing)
- Output: prediction + probability + short explanation of which words/phrases influenced the result

Key terms

- NLP (Natural Language Processing): making a computer understand and work with human language (emails/messages).
- Supervised ML: training with examples that already have correct answers (labels).
- Label: the correct answer column (0 = legit, 1 = phishing).
- Feature: numeric representation of text that a model can learn from.
- TF-IDF: converts text into numbers; gives higher weight to words/phrases that are more informative.
- n-grams: groups of words:
 - Unigram: 1 word (“verify”)
 - Bigram: 2 words (“verify account”)
 - Trigram: 3 words (“verify your account”)
- Logistic Regression: a simple, strong classifier that outputs a probability (phishing likelihood).
- C value (Logistic Regression): controls flexibility
 - smaller C = simpler/more cautious model
 - larger C = more flexible model (can overfit if too high)

The 6-step demo workflow

Step 1 : Load dataset (CSV)

- Upload/select the training CSV in Colab
- Create one input column: text = subject + body

Step 2 : Train/Test split

- Train set: the model learns from these emails
- Test set: unseen emails used for fair evaluation

Step 3 : Build the pipeline

- TF-IDF (text → numbers) → Logistic Regression (numbers → probability/label)

Step 4 : Train

- Train the pipeline on the training set

Step 5 : Evaluate (metrics + confusion matrix)

- Accuracy: overall correctness
- Precision: when it says “phishing”, how often it’s right (false alarms)
- Recall: out of all phishing emails, how many it catches (missed attacks)
- Confusion Matrix: visual table of TP/TN/FP/FN

Step 6 : Predict + Explain + Interaction

- Paste/type any email → PHISHING/LEGIT + probability
- “WOW” explanation: show the top n-grams pushing toward phishing vs legit
- Change n-grams and C → retrain → see how predictions and metrics change
- Use sample dropdown + filter (ALL/LEGIT/PHISHING) for quick practice

How to interpret results quickly

- Low recall → model misses phishing (risky)
- Low precision → too many false alarms (annoying)
- Confusion matrix shows where errors are happening (FP vs FN)

Reading resource (Cybersecurity + NLP)

- Proofpoint (overview): <https://www.proofpoint.com/au/threat-reference/natural-language-processing>
- Optional alternative (more engineering): <https://developer.nvidia.com/blog/changing-cybersecurity-with-natural-language-processing/>

Canadian cyber networks and official guidance

- Canadian Centre for Cyber Security (official): <https://www.cyber.gc.ca/en>
- Practical advisories & guidance: <https://www.cyber.gc.ca/en/guidance>
- Canadian Cybersecurity Network (ecosystem/community): <https://canadiancybersecuritynetwork.com/>
- National Cybersecurity Consortium (pan-Canadian network): <https://ncc-cnc.ca/>

Co-ops / entry-level cybersecurity links (Canada)

- LinkedIn (Cyber Security Intern roles): <https://ca.linkedin.com/jobs/cyber-security-intern-jobs>
- Indeed (Cyber Security Co-op roles): <https://ca.indeed.com/q-cyber-security-co-op-jobs.html>
- Prosple (Internships): <https://ca.prosple.com/cyber-security-internships-canada>

Cybersecurity job link (Canada)

- Dedicated portal: <https://canadiancybersecurityjobs.com/>

Official current report (Canada)

- Cyber Centre: Ransomware Threat Outlook 2025–2027 (Canada.ca): <https://www.canada.ca/en/communications-security/news/2025/12/cyber-centre-releases-ransomware-threat-outlook-2025-to-2027.html>